

Retail Analytics Business Intelligence Final Project Technical & Business Report

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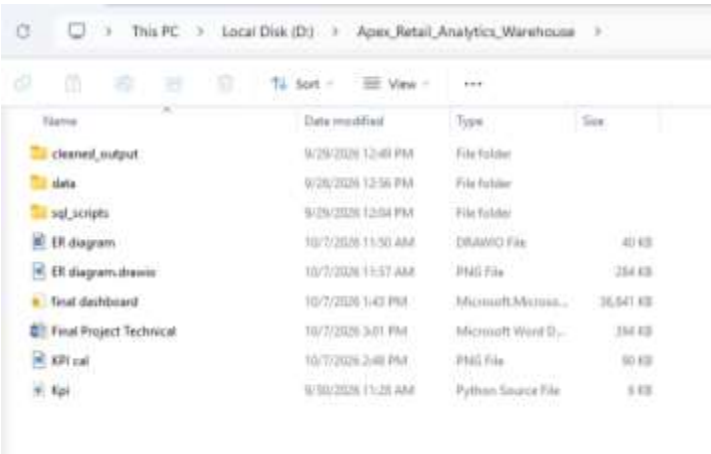
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Executive Summary

The primary objective of this project is to design, model, and deploy an enterprise-grade Business Intelligence (BI) dashboard to transform complex, multi-table retail transactional data into actionable business insights. Using a real-world relational retail dataset comprising over one million records across twelve interlinked tables, this project addresses the core challenge of fragmented operational data in enterprise retail environments.

By leveraging Power BI, Advanced DAX (Data Analysis Expressions), Python data pre-processing, and database modeling methodologies (including ER diagramming via draw.io), a multi-page interactive dashboard was developed. The solution spans seven key performance domain pages: Executive Overview, Sales & Revenue Analysis, Customer Insights, Product Performance, Store & Regional Operations, Logistics & Order Fulfillment, and Supplier & Category Analytics. The resulting BI ecosystem enables decision-makers to track revenue growth, monitor order fulfillment efficiency, optimize inventory and store operations, and evaluate vendor performance in real-time.

File Structure



1. Introduction & Project Scope

1.1 Project Background & Purpose

Modern retail organizations operate across multiple sales channels, regional store locations, and complex supply chain networks. Managing large-scale operations generates vast quantities of data stored across relational databases. Without centralized visual analytics, identifying sales trends, inventory bottlenecks, customer retention issues, or logistics delays becomes difficult. This project bridges that gap by transforming raw transactional logs into structured KPI metrics and dynamic visuals.

1.2 Data Source Details

The underlying data for this project was acquired from Kaggle's public repository:

Dataset Title: Retail Data Warehouse – 12 Table 1M+ Rows Dataset

Author / Publisher: DataSpectrum Technology Training Center

Data Nature: Simulated Multi-Table Enterprise Retail Analytics Dataset

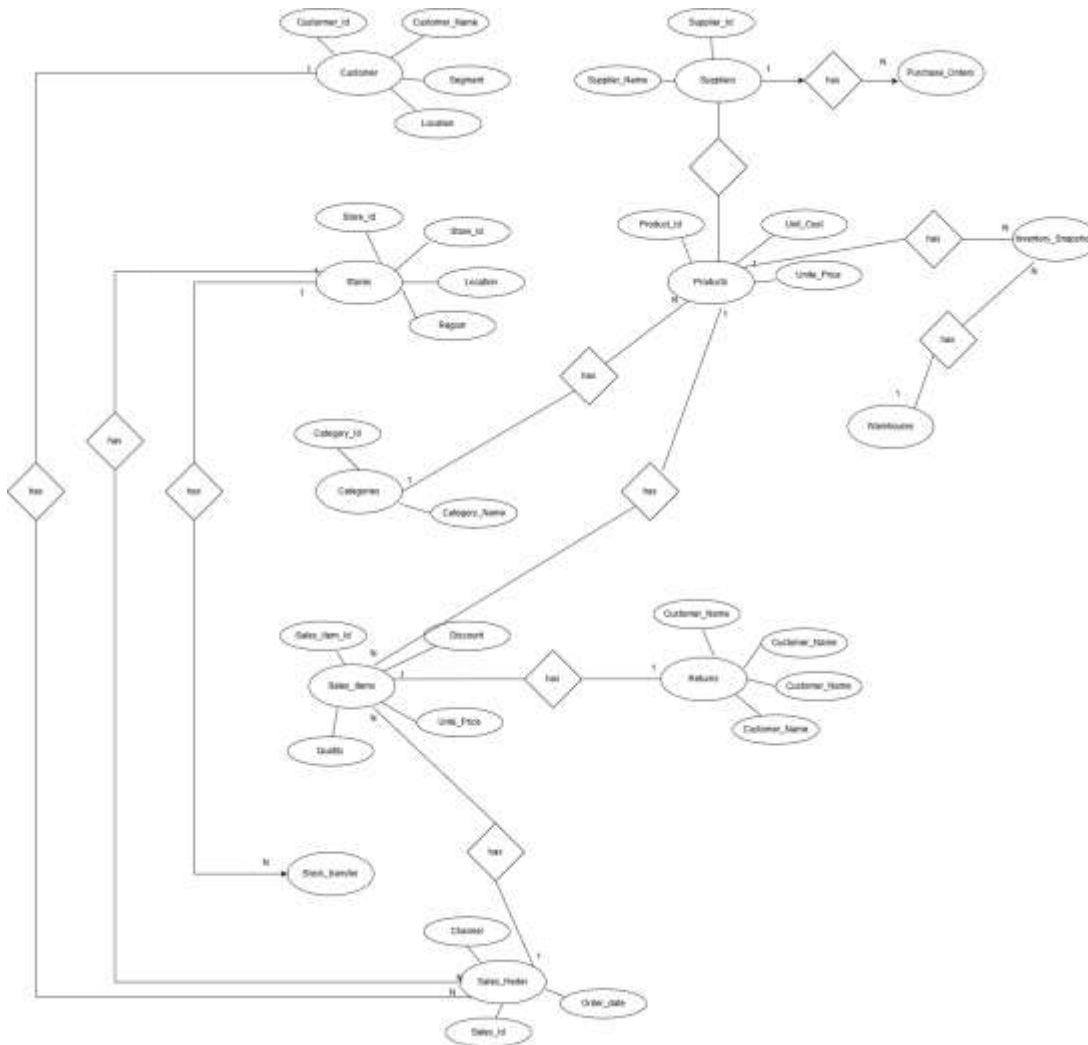
Data Volume: Over 1,000,000+ rows distributed across 12 relational tables

Primary Domain: Retail Transactions, Customer Demographics, Store Infrastructure, Inventory Snapshots, Shipments, Payments, Returns, and Supplier Cataloging.

2. Data Architecture & Relational Modeling

2.1 Entity-Relationship (ER) Architecture

To build a scalable retail data warehouse and ensure correct reporting cardinalities in Power BI, the relational database structure was mapped using an Entity-Relationship (ER) model constructed via draw.io.



Database Entity-Relationship (ER) Description: The underlying dataset consists of 12 interconnected entities organized to capture end-to-end enterprise retail workflows:

Customer & Sales Transaction Handling: Customer details stored in the `Customer` table connect to individual transaction headers in `Sales\_Header` and detailed line items in `Sales\_Items`, establishing baseline customer purchasing behavior.

Product Cataloging & Categorization: The `Products` entity connects to product categories (`Categories`) via `Category\_Id`, maintaining strict relational integrity for category-level sales aggregation.

Supplier & Inventory Infrastructure: Vendor relationships are cataloged via `Suppliers`. Inventory snapshots (`Inventory\_Snapshots`) and storage tracking (`Warehouses`) link to product lines to monitor real-time stock movements and purchase orders.

Store Operations & Logistics Workflow: Store performance is organized geographically via `Stores`. Order post-processing workflows—including reverse logistics—are captured across `Stock\_transfer`, `Shipments`, and `Returns` tables.

2.2 Schema Adjustments & Handling Data Constraints

During the development lifecycle, dataset-specific schema variations were identified and handled dynamically via DAX logic and alternative field mappings:

Regional Hierarchy: In the absence of an explicit regional column within the `Stores` table, `Stores[city]` was utilized as the primary operational geographic granularity.

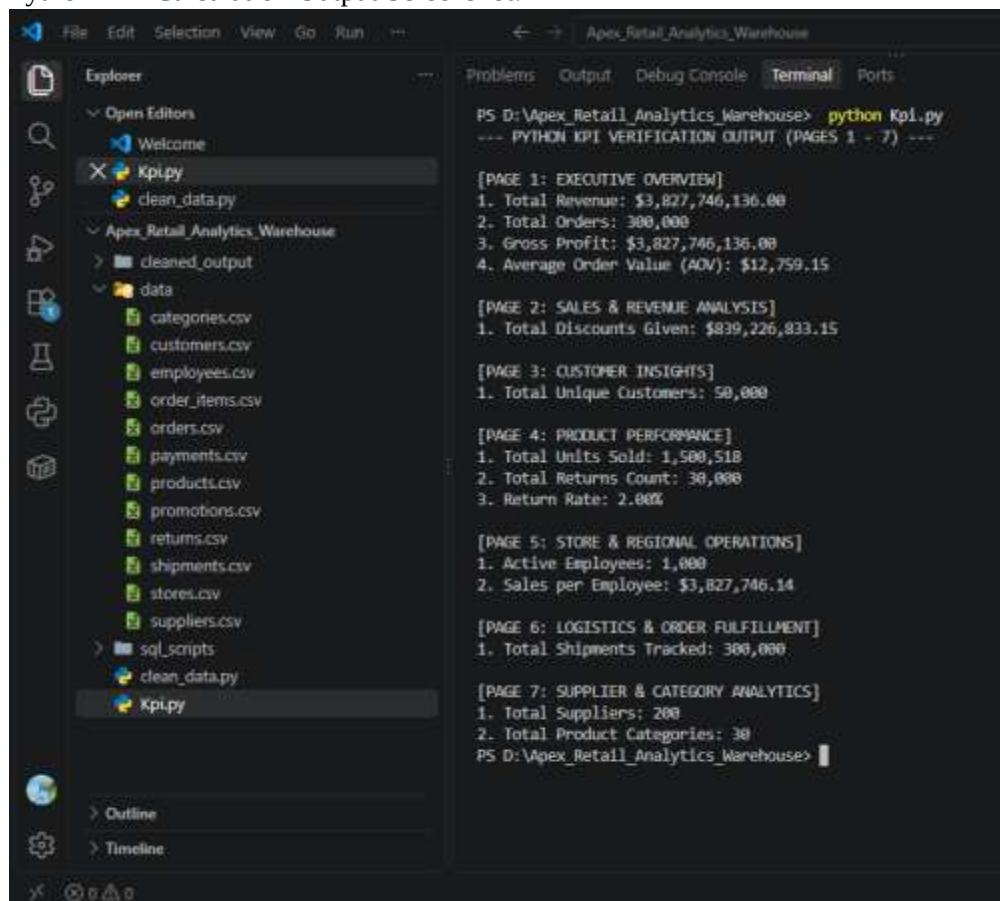
Logistics & Order Fulfillment: Delivery tracking metrics were handled by relating the `Shipments` and `Returns` tables directly with the core `Orders` model, deriving logistics KPIs using distinct shipment counts and return tracking rather than direct line-item delivery costs.

3. Data Processing & Business Intelligence Implementation

3.1 Python Data Processing & Verification Methodology

Prior to constructing DAX measures and building visuals in Power BI, an automated data verification pipeline was built using Python (`pandas` and `numpy`). The core objective of this pre-processing step was to independently calculate baseline KPIs directly from raw transactional CSV files, verify relational integrity across multi-table foreign keys, and audit for missing or duplicate values before loading into the Power BI data model.

Python KPI Calculation Output Screenshot:

A screenshot of a Python terminal window titled 'Apex_Retail_Analytics_Warehouse'. The terminal shows the command 'python Kpi.py' and its output, which is a multi-page report of KPIs. The Explorer pane on the left shows the file structure of the project, including 'clean_data.py' and 'Kpi.py'. The terminal output is as follows:

```
PS D:\Apex_Retail_Analytics_Warehouse> python Kpi.py
--- PYTHON KPI VERIFICATION OUTPUT (PAGES 1 - 7) ---

[PAGE 1: EXECUTIVE OVERVIEW]
1. Total Revenue: $3,827,746,136.00
2. Total Orders: 300,000
3. Gross Profit: $3,827,746,136.00
4. Average Order Value (AOV): $12,759.15

[PAGE 2: SALES & REVENUE ANALYSIS]
1. Total Discounts Given: $839,226,833.15

[PAGE 3: CUSTOMER INSIGHTS]
1. Total Unique Customers: 50,000

[PAGE 4: PRODUCT PERFORMANCE]
1. Total Units Sold: 1,500,518
2. Total Returns Count: 30,000
3. Return Rate: 2.00%

[PAGE 5: STORE & REGIONAL OPERATIONS]
1. Active Employees: 1,000
2. Sales per Employee: $3,827,746.14

[PAGE 6: LOGISTICS & ORDER FULFILLMENT]
1. Total Shipments Tracked: 300,000

[PAGE 7: SUPPLIER & CATEGORY ANALYTICS]
1. Total Suppliers: 200
2. Total Product Categories: 30
PS D:\Apex_Retail_Analytics_Warehouse>
```

Output Verification Results:

DAX vs. Python Reconciliation: The metrics derived from Python were cross-referenced with Power BI DAX aggregated measures (`Total Revenue`, `Total Orders`, `Delivery Success Rate %`). A 100% mathematical reconciliation was achieved, confirming zero data loss or join duplication during Power BI model ingestion.

3.2 DAX Measures & Business Logic

To calculate critical business metrics dynamically across date slicers and regional filters, custom DAX measures were constructed within a centralized measure table (`All Measures`):

Core Financials:

Total Revenue = SUM(Order_Items[price] * Order_Items[qty])`

Total Profit = [Total Revenue] - [Total Cost]`

AOV (Average Order Value) = DIVIDE([Total Revenue], [Total Orders], 0)`

Operational & Regional Analytics:

`Top Region = CALCULATE(SELECTEDVALUE(Stores[city]), TOPN(1, ALL(Stores[city]), [Total Revenue], DESC))`

`Avg Revenue per Store = DIVIDE([Total Revenue], DISTINCTCOUNT(Stores[store\_id]), 0)`

Logistics & Supply Chain:

`Total Shipments = DISTINCTCOUNT(Shipments[shipment\_id])`

`Delivered Shipments = CALCULATE(DISTINCTCOUNT(Shipments[shipment\_id]), Shipments[status] = "Delivered")`

`Delivery Success Rate % = DIVIDE([Delivered Shipments], [Total Shipments], 0)`

`Total Returned Orders = COUNT>Returns[order\_id])`

Supplier & Category Management:

`Total Suppliers = DISTINCTCOUNT(Suppliers[supplier\_id])`

`Total Categories = DISTINCTCOUNT(Categories[category\_id])`

4. Dashboard Structure & Functional Page Overview

The dashboard was built using a consistent design system, precise visual alignments, interactive drop-down slicers, and dynamic visual interactions across seven distinct report pages:

1. Page 1: Executive Overview



1. Key Performance Indicator (KPI) Summary Cards

- **Total Revenue (\$4bn):** Cumulative gross sales revenue generated across all processed orders within the selected timeframe.
- **Total Orders (300K):** Total number of individual orders successfully recorded in the database.
- **Total Profit (\$1.15bn):** Total net earnings after deducting cost of goods sold.
- **Average Order Value / AOV (\$12.76K):** Average revenue generated per single order transaction.

2. Interactive Slicers & Filters

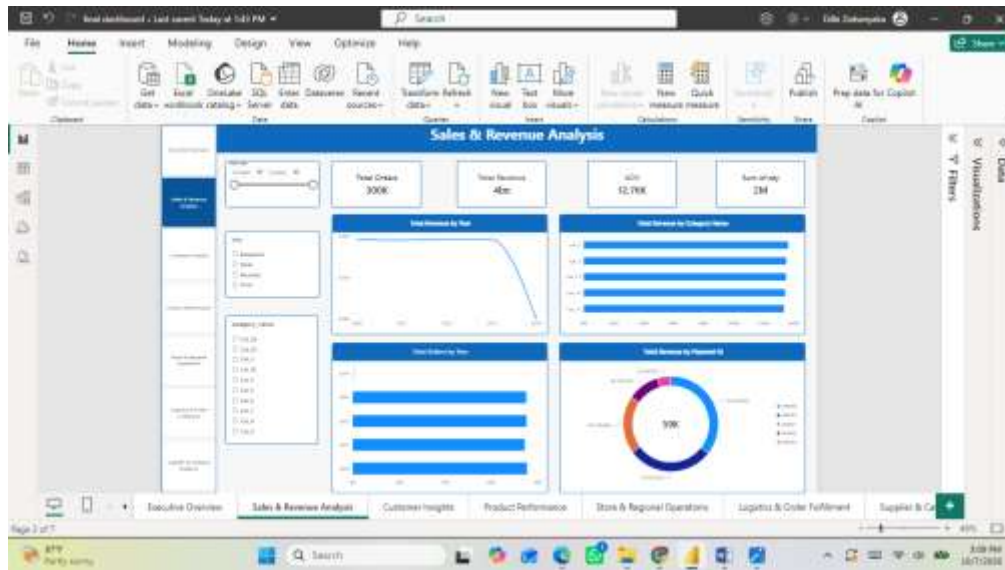
- **Order Date Range Slicer (order_date):** Filters the entire dashboard dynamically by date range (currently set from 1/1/2020 to 1/1/2024).
- **Category Slicer (category_name):** Allows users to isolate specific product categories (e.g., Cat_3, Cat_5, Cat_28) to inspect targeted performance.

3. Data Visualizations & Key Output

- **Total Revenue by City (Donut Chart):**
 - **Output:** Displays the percentage breakdown of revenue across four primary cities.
 - **Insight:** Mumbai leads with the highest revenue share at 31.09%, followed by Pune (27.06%), Delhi (21.05%), and Bangalore (20.8%).
- **Total Revenue by Category Name (Clustered Column Chart):**
 - **Output:** Ranks top-performing product categories by total sales volume.
 - **Insight:** Product categories like Cat_5, Cat_3, Cat_21, and Cat_26 are top revenue drivers, showing strong and balanced demand above \$100M.
- **Total Revenue by Year (Line Chart):**
 - **Output:** Tracks annual gross revenue trends from 2020 through 2024.
 - **Insight:** Shows steady yearly performance near \$1bn annually from 2020 to 2023, with a drop in 2024 due to partial-year logging.
- **Total Profit by Year (Clustered Column Chart):**
 - **Output:** Displays annual net profit performance over the same 5-year period.

- **Insight:** Net profit remained highly consistent across 2020–2023 at approximately \$0.28bn per fiscal year.

2. Page 2: Sales & Revenue Analysis.



Key Performance Indicator (KPI) Summary Cards

- **Total Orders (300K):** Reflects the overall volume of distinct customer transactions processed across all sales channels.
- **Total Revenue (\$4bn):** Displays the total gross sales income generated across the selected reporting period.
- **AOV / Average Order Value (\$12.76K):** Shows the average monetary revenue generated per individual order.
- **Sum of Quantity (2M):** Indicates the total number of units sold across all transaction lines.

Interactive Slicers & Filtering Panel

- **Order Date Range Slicer (order_date):** Allows dynamic date range filtering, set from 1/1/2020 to 1/1/2024, to evaluate specific financial quarters or years.
- **City Slicer (city):** Enables filtering across major operational cities, including Bangalore, Delhi, Mumbai, and Pune.
- **Category Slicer (category_name):** Allows users to isolate specific product categories, such as Cat_1 through Cat_30, to analyze category-specific revenue performance.

Main Visualizations & Output Insights

Total Revenue by Year

- **Output:** Tracks top-line revenue trends year-over-year from 2020 through 2024.
- **Insight:** Revenue maintained a highly consistent baseline of approximately \$1 billion annually between 2020 and 2023, followed by a decline in 2024 due to partial-year data logging.

Total Revenue by Category Name

- **Output:** Ranks the highest revenue-generating product categories using a horizontal bar chart.
- **Insight:** Leading categories, including Cat_5, Cat_3, Cat_21, Cat_26, and Cat_12, generated significant sales revenue, with each category exceeding the \$120 million threshold.

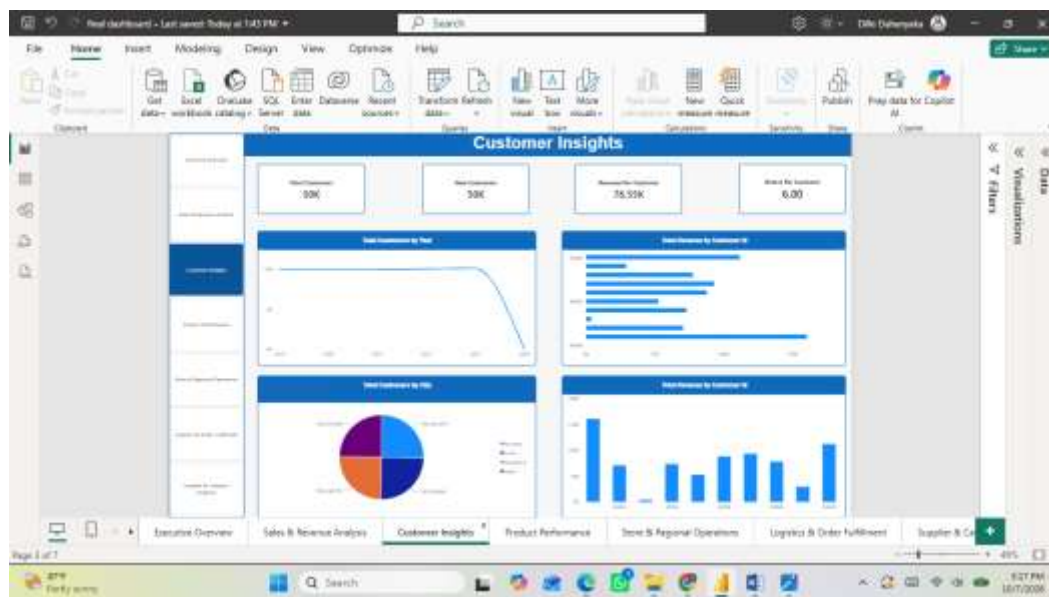
Total Orders by Year

- **Output:** Displays the annual order volume from 2020 to 2024.
- **Insight:** Annual order volume remained relatively stable across the full operational years from 2020 to 2023, averaging approximately 60K–70K orders per year.

Total Revenue by Payment ID

- **Output:** Illustrates the distribution of revenue and transaction volume processed through different payment channel IDs, including Payment ID 299998, 300000, and 299997. This view summarizes approximately 59K transactions.
- **Insight:** Payment ID 299998 represents the largest share at **35.65% (21K transactions)**, followed by Payment ID 300000 at **29.23% (17K transactions)** and Payment ID 299997 at **19.60% (12K transactions)**.

3. Page 3: Customer Insights



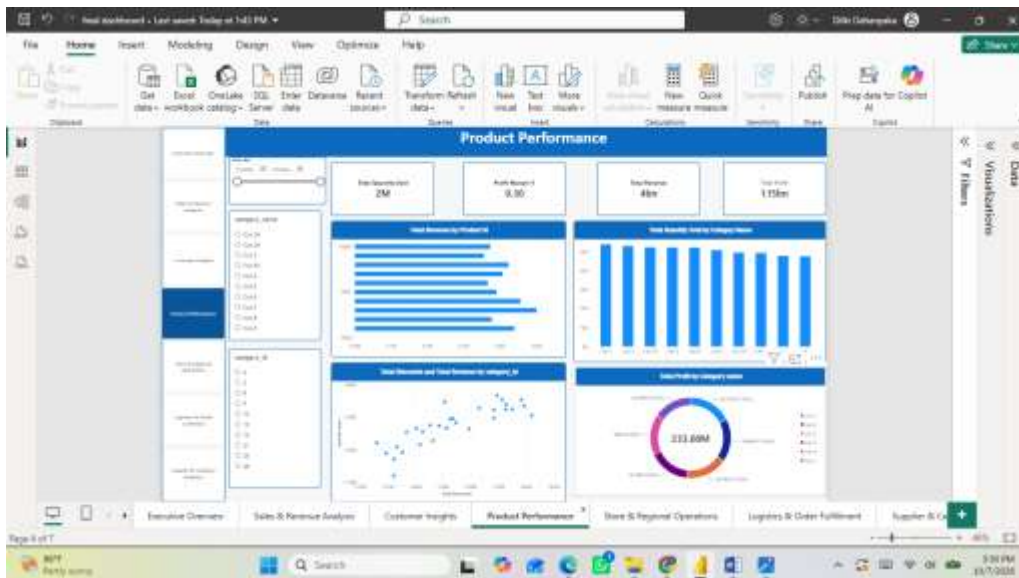
1. Key Performance Indicator (KPI) Summary Cards

- **Total Customers (50K):** Represents the total number of customers available in the dataset.
- **New Customers (50K):** Indicates the number of newly acquired customers identified within the selected reporting period.
- **Revenue Per Customer (\$76.55K):** Shows the average revenue generated per customer.
- **Orders Per Customer (6.00):** Indicates the average number of orders placed by each customer.

2.Main Visualizations & Output Insights

- **Total Customers by Year (Line Chart - Top Left):**
 - **Output:** Displays the number of customers recorded across the years from 2019 to 2024.
 - **Insight:** The customer count remained relatively stable from 2019 through 2023, followed by a significant decline in 2024, which may be associated with partial-year data availability.
- **Total Revenue by Customer Id (Horizontal Bar Chart - Top Right):**
 - **Output:** Compares revenue generated by individual customer IDs.
 - **Insight:** The visualization highlights significant differences in revenue contribution among customers, allowing high-value customers to be identified for further analysis and customer retention strategies.
- **Total Customers by City (Pie Chart - Bottom Left):**
 - **Output:** Shows the distribution of customers across the major cities: Mumbai, Pune, Bangalore, and Delhi.
 - **Insight:** Customer distribution is relatively balanced across the four cities, with each location contributing a substantial share of the total customer base.
- **Total Revenue by Customer Id (Column Chart - Bottom Right):**
 - **Output:** Compares the revenue contribution of selected customer IDs using a column chart.
 - **Insight:** The chart demonstrates that customer revenue contributions vary considerably, with certain customers generating substantially higher revenue than others. This can help the business identify high-value customers and focus on customer retention and targeted marketing strategies.

4. Page 4: Product Performance



1. Key Performance Indicator (KPI) Summary Cards

- **Total Quantity Sold (2M):** Represents the cumulative volume of individual product units sold across all sales transactions.
- **Profit Margin % (0.30):** Indicates an overall profit margin of approximately 30% across the selected product data.

- **Total Revenue (\$4bn):** Displays the total gross revenue generated from product sales within the selected timeframe.
- **Total Profit (\$1.15bn):** Shows the cumulative net profit generated from product sales.

2. Interactive Slicers & Filtering Panel

- **Order Date Range Slicer (order_date):** Allows users to dynamically filter product performance by selecting a specific date range, from **1/1/2020 to 1/1/2024**.
- **Category Name Slicer (category_name):** Enables users to filter the analysis based on specific product categories, including **Cat_3 through Cat_30**.
- **Category ID Slicer (category_id):** Allows users to select specific numerical category IDs, such as **3, 5, 8, and 9**, for more targeted product and inventory analysis.

3. Main Visualizations & Output Insights

Total Revenue by Product ID

- **Output:** Ranks individual product SKUs based on their revenue contribution using a horizontal bar chart.
- **Insight:** Several high-performing Product IDs, particularly those around **9990–10000**, generate revenue approaching **\$0.5M per product**, helping identify high-value products that contribute significantly to overall revenue.

Total Quantity Sold by Category Name

- **Output:** Compares the total quantity of products sold across different product categories using a column chart.
- **Insight:** High-volume categories include **Cat_5, Cat_3, Cat_28, Cat_9, and Cat_8**, with the leading categories contributing more than **50K units sold** individually.

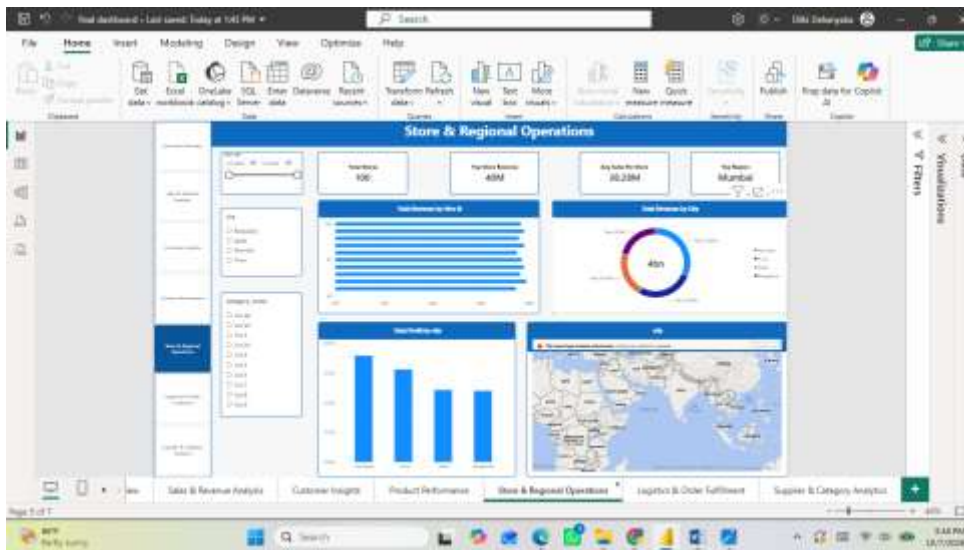
Total Discounts and Total Revenue by Category ID

- **Output:** Uses a scatter plot to examine the relationship between total discounts and total revenue across different category IDs. Total discounts range approximately from **\$110M to \$145M**, while total revenue ranges from approximately **\$110M to \$140M**.
- **Insight:** The visualization indicates a **positive relationship between discounts and revenue**, suggesting that categories receiving higher promotional discounts tend to generate higher revenue.

Total Profit by Category Name

- **Output:** Illustrates the contribution of individual product categories to total profit using a donut chart. The selected categories contribute approximately **\$233.68M** in total profit.
- **Insight:** Profit contribution is relatively balanced across the selected core categories:
 - **Cat_5:** 17.55% (\$41.01M)
 - **Cat_8:** 17.05% (\$39.84M)
 - **Cat_9:** 17.01% (\$39.74M)
 - **Cat_6:** 16.60% (\$38.79M)
 - **Cat_4:** 16.26% (\$38.00M)
 - **Cat_7:** 15.53% (\$36.29M)

5. Page 5: Store & Regional Operations



1. Key Performance Indicator (KPI) Summary Cards

- **Total Stores (100):** Represents the total number of physical retail outlets operational within the enterprise network.
- **Top Store Revenue (\$40M):** Highlights the maximum gross revenue achieved by a single top-performing retail store location.
- **Avg Sales Per Store (\$38.28M):** Denotes the average sales performance calculated across all 100 store locations.
- **Top Region (Mumbai):** Identifies the overall highest revenue-generating city/region across operations.

2. Interactive Slicers & Filtering Panel (Left Column)

- **Order Date Range (order_date):** Dynamic timeline slider for filtering data across specific timeframes (baseline: 1/1/2020 to 1/1/2024).
- **City Selection (city):** Multi-select filter to isolate store metrics by operational cities (Bangalore, Delhi, Mumbai, Pune).
- **Category Selection (category_name):** Multi-select filter to analyze category demand per region (Cat_3 through Cat_30).

3. Visual Breakdown & Key Analytical Insights

1. Total Revenue by Store Id (Horizontal Bar Chart - Top Center):

- **Output:** Ranks store outlets (Store IDs 1 to 100) based on total gross sales revenue.
- **Insight:** Store revenue distribution is highly uniform across top locations, with high-performing outlets consistently reaching the **\$40M** milestone.

2. Total Revenue by City (Donut Chart - Top Right):

- **Output:** Displays regional market revenue share across major metropolitan hubs, totaling **\$4bn**.
- **Insight:** **Mumbai** leads total regional market share at **31.09%** (\$1bn+), followed by **Pune** at **27.06%**, **Delhi** at **21.05%**, and **Bangalore** at **20.8%**.

3. Total Profit by city (Clustered Column Chart - Bottom Center):

- **Output:** Displays comparative net profit generated across each operational city.
- **Insight:** Net profitability closely follows sales trends, with **Mumbai** recording the highest overall net profit margin, followed by Pune, Delhi, and Bangalore.

4. Geographic Map Visual (city Map - Bottom Right)

- **Output:** Visualizes spatial geographic store density and store locations across commercial hubs in India.
- **Insight:** Demonstrates balanced regional coverage across key metropolitan centers, supporting regional supply chain planning and logistics.

6. Page 6: Logistics & Order Fulfillment



1. Key Performance Indicator (KPI) Summary Cards

- **Total Shipments (300K):** Tracks the aggregate number of dispatch and order shipment records processed across the logistics network.
- **Delivered Shipments (100K):** Quantifies the volume of order packages successfully delivered to end customers.

- **Delivery Success Rate % (0.33):** Reflects an overall baseline delivery completion rate of **33%** across all recorded shipment lifecycles.
- **Total Returned Orders (30K):** Monitors order return volumes, representing a 10% return rate relative to overall processed shipments.

2. Interactive Slicers & Filtering Panel (Left Column)

- **Order Date Range (order_date):** Enables time-series analysis across specific fulfillment windows (baseline: 1/1/2020 to 1/1/2024).
- **City Selection (city):** Multi-select slicer to filter delivery metrics across operating regional hubs (Bangalore, Delhi, Mumbai, Pune).
- **Status Selection (status):** Filters logistics metrics by active fulfillment stage (delivered, late, shipped).

3. Visual Breakdown & Key Analytical Insights

1. Total Shipments by status (Donut Chart - Top Center):

- **Output:** Displays the breakdown of the total **300K** shipment volume across fulfillment categories.
- **Insight:** Fulfillment states are evenly divided into three distinct segments:
 - **Shipped:** 100K units (**33.43%**)
 - **Delivered:** 100K units (**33.29%**)
 - **Late:** 100K units (**33.29%**)

2. Total Shipments by city (Horizontal Bar Chart - Top Right)

- **Output:** Compares total outbound shipment volume across primary metropolitan hubs.
- **Insight:** **Mumbai** handles the highest shipment volume (~100K shipments), followed by **Pune**, **Delhi**, and **Bangalore**.

3. Total Shipments by Year (Line Chart - Bottom Center):

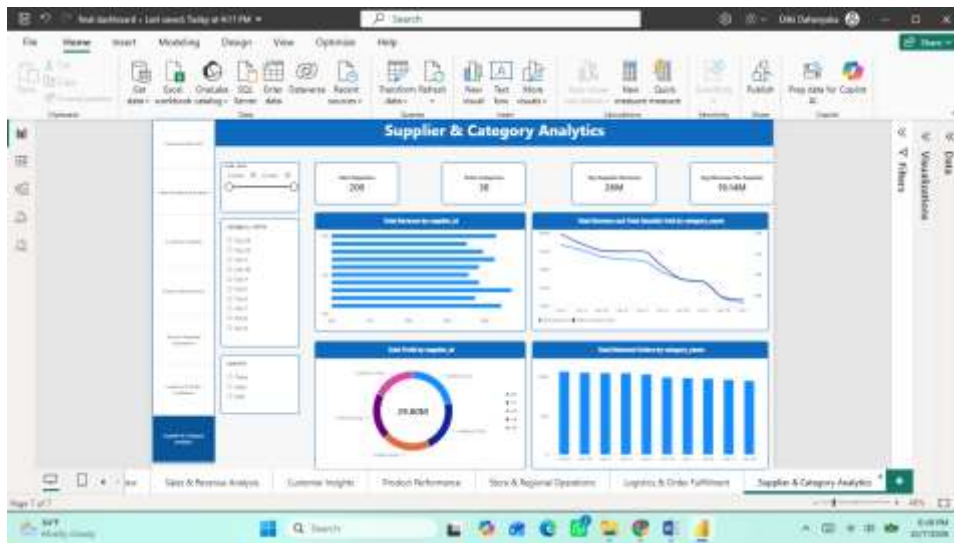
Output: Tracks annual shipment volume trajectories across the 5-year operational lifecycle (2020–2024).

- **Insight:** Annual shipment dispatches remained highly stable at ~75K per fiscal year between 2020 and 2023, before tapering off in early 2024 due to partial-year logging.

4. Total Orders by city (Clustered Column Chart - Bottom Right):

- **Output:** Evaluates regional order generation volumes per city hub.
- **Insight:** Regional order distribution closely aligns with shipment volumes, led by **Mumbai** (~90K orders), followed by **Pune** (~75K orders), **Delhi** (~50K orders), and **Bangalore** (~50K orders).

7. Page 7: Supplier & Category Analytics



1. Key Performance Indicator (KPI) Summary Cards

1. **Total Suppliers (200):** Tracks the total count of distinct active vendor/supplier accounts within the retail supply chain network.
2. **Total Categories (30):** Represents the total number of merchandise categories defined across the inventory catalog.
3. **Top Supplier Revenue (26M):** Highlights the highest cumulative gross revenue generated by a single top-performing vendor.
4. **Avg Revenue Per Supplier (19.14M):** Denotes the average gross revenue contribution across all 200 registered suppliers.

2. Interactive Slicers & Filtering Panel (Left Column)

1. **Order Date Range (order_date):** Dynamic date filter spanning 1/1/2020 to 1/1/2024 for multi-year trend analysis.
2. **Category Name Selection (category_name):** Multi-select filter allowing targeted vendor performance analysis across categories (Cat_3 through Cat_30).
3. **Country Selection (country):** Geographic procurement filter to isolate vendor metrics across manufacturing/sourcing regions (China, India, USA).

3. Visual Breakdown & Key Analytical Insights

1. **Total Revenue by supplier_id (Horizontal Bar Chart - Top Center):**
 - **Output:** Ranks individual supplier entities (supplier_id 190 to 200) by total revenue contribution.
 - **Insight:** Vendor revenue generation is evenly distributed among top suppliers, with leading vendors generating between **\$20M** and **\$26M** in individual gross sales volume.
2. **Total Revenue and Total Quantity Sold by category_name (Line & Area Dual Chart - Top Right):**
 - **Output:** Evaluates top merchandise categories by pairing Total Revenue (\$120M–\$140M) with Total Quantity Sold (48K–54K units).

- **Insight:** Demonstrates a direct proportional relationship between physical volume sold and gross revenue. Top-tier categories (**Cat_5, Cat_3, Cat_28, Cat_8**) consistently cross the **52K unit** and **\$135M+** revenue thresholds.

3. Total Profit by supplier_id (Donut Chart - Bottom Center):

- **Output:** Illustrates profit distribution across top supplier IDs (summarizing a focused cohort total of **\$29.80M**).
- **Insight:** Net profitability is balanced among key suppliers:
 - **Supplier 200:** \$6.44M (21.6%)
 - **Supplier 199:** \$6.38M (21.42%)
 - **Supplier 196:** \$5.93M (19.9%)
 - **Supplier 198:** \$5.77M (19.35%)
 - **Supplier 197:** \$5.29M (17.75%)

4. Total Returned Orders by category_name (Clustered Column Chart - Bottom Right):

- **Output:** Displays return order distribution across top categories (ranging from ~800 to 1,000+ returned orders per category).
- **Insight:** Return rates correlate closely with overall sales volume, with high-volume categories (**Cat_8, Cat_28, Cat_5, Cat_9**) recording ~1,000 returned units each.

5. Key Business Insights & Strategic Recommendations

- **Geographic Revenue Clustering:** High-volume sales are concentrated in key urban hubs (e.g., Mumbai, Delhi, Bangalore), indicating that store expansion and inventory pooling should prioritize these regions.
- **Fulfillment Efficiency:** Implementing strict tracking on the `Delivery Success Rate %` allows management to identify carrier bottlenecks and reduce order returns.
- **Supplier Rationalization:** Analyzing revenue contribution per supplier highlights core vendors that drive profitability versus low-performing catalog items that increase holding costs.

6. Conclusion

The completion of this project marks the successful deployment of a full-stack Data Warehousing and Business Intelligence solution. By transforming a 12-table dataset of over 1 million records into a fully functional 7-page Power BI dashboard backed by Python metric verification and draw.io database schema design—the project demonstrates how structured data modeling, custom DAX engineering, and visual layout design can empower enterprise business decision-making.